OpenMobile-2: Building Versatile Mobile Agents with Scalable Environments and App-Native Tools

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Paper version: arXiv 2610.14157v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *reproducing rich workflows and persistent user states at scale has remained a key obstacle.*

Open agents train on **few, shallow apps** — commercial workflows need reset, state, and verification at scale.

Motivation: three open gaps

1. **Emulator apps:** realistic, resist clean reset.
2. **Simulated apps:** cheap reset, shallow flows.
3. **GUI-only agents:** miss app-native tools.

Evaluation setup

AndroidWorld · 116 | MobileWorld · 201 | MobileGym++ Bench · 215 | SPA-Bench L3 (real) | Qwen3.5-9B + 27B bases

- **All online** success rates; no offline step accuracy.
- Emulator, simulator, **real-device** suites; SFT then RL.

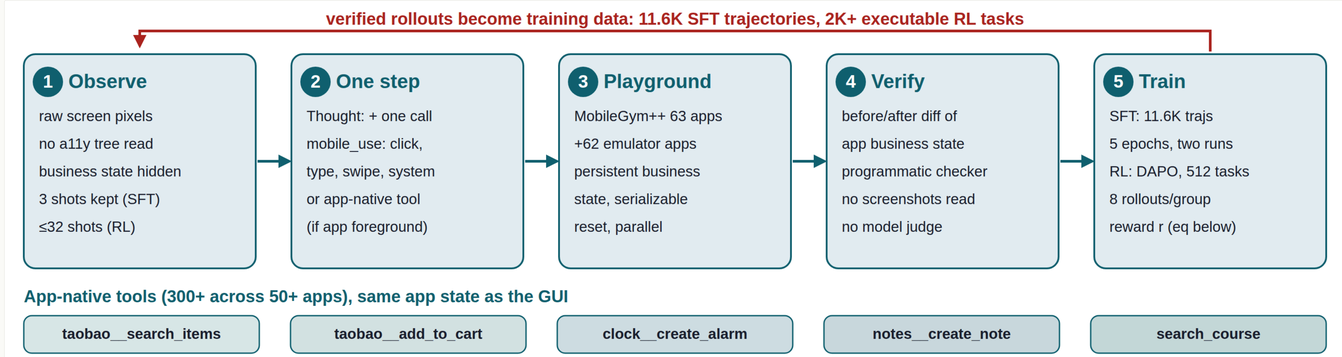
Our read (CUA Radar)

62	★★	transfer
score of 100	band	evidence level

- **Reports:** online eval on 5 suites; ± std over eval seeds; strong baselines; coverage ablation; overlap audit; released code.
- **Flags:** AP2, AP5; held-out apps only partial (74 InD / 42 held-out); unvalidated VLM judge on emulator SFT; budget unclear.
- Pattern C22: resettable apps + state checkers; authors' numbers, not rerun.

Method: a verified commercial playground feeding SFT and RL

- **Sees / acts:** pixels only (no a11y tree); mobile_use click / type / swipe at 999×999, or an app-native tool when its app is foreground.
- **Playground:** 35 new commercial replicas (63 sim clients) + 62 emulator apps; persistent business state, serializable reset.
- **Signal:** programmatic checker on the business-state diff; SFT on 11.6K trajectories, then RL with the reward below.



Our redraw (not the paper's figure): one OpenMobile-2 step from screenshot to verified business-state write, and the data loop it feeds.

$$r = v \cdot m_{\text{tool}} \cdot \ell \; - \; 0.2\, n_{\text{fmt}} - 0.2\, f_{\text{succ}}$$

RL reward (App. A.2): relaxed verifier × tool bonus × length factor − format / false-success penalties

Key results: online success rate (%)

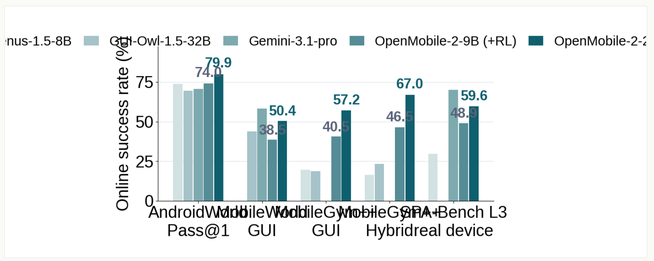


Table 2, redrawn: online success (%), final checkpoints; ± std where reported.

SPA-Bench L3 real: 31.9 (base) → 59.6 (27B+RL)

SO WHAT → *RL gains concentrate in aligned envs (MobileGym SR 48.4→51.2, 27B).*

Ablation: 27B training stages

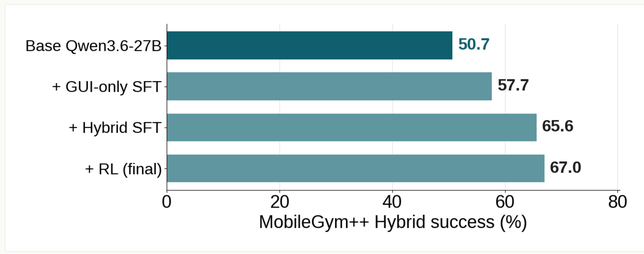


Table 2, redrawn: MobileGym++ Hybrid success (%) at each stage of the 27B pipeline.

SO WHAT → *SFT on the open data drives the main gain; hybrid tool data lifts Hybrid eval from 57.7 to 65.6, RL from 65.6 to 67.0. The 9B model shows the same order: 12.1 (base), 41.9 (GUI-only SFT), 44.7 (hybrid), 46.5 (+RL). Scaling app coverage from 15 to 53 at a fixed 2K budget lifts held-out AndroidWorld from 60.7 to 67.5 while InD stays 73.7 to 71.4.*

Headline Numbers

79.9
AndroidWorld
Pass@1; 27B +RL (base 67.1)

- **59.6** real-device SPA-Bench
- **67.0** MobileGym++ Hybrid
- **11.6K** open trajectories

The paper's own controls

CONTROL	RESULT	READING
Goal overlap	≤2.25% above 0.7	none above 0.9
Coverage 15→53	held-out 60.7→67.5	InD 73.7→71.4
Tools + hybrid SFT	gain 5.6 / 10.3 pts	vs 2.8–4.2 without
Unseen tools (MCP)	27.5 vs 12.5 (9B)	45.0 vs 35.0 (27B)

Coverage → held-out gains; tools need hybrid SFT

Takeaway

110+ apps, 11.6K trajectories and 2K+ RL tasks lift Qwen bases to 79.9 AndroidWorld and 59.6 real-device SPA-Bench.

35 commercial replicas | 300+ app-native tools | MobileGym++ Bench · 215

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